

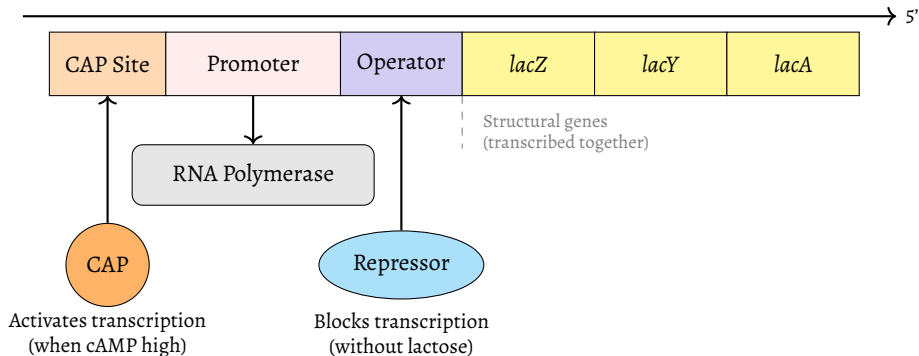


Figure 12 – Schematic representation of the *lac* operon in *E. coli*. The CAP site activates transcription when cAMP levels are elevated (low glucose). The LacI repressor binds to the operator in the absence of lactose, physically blocking RNA polymerase and preventing the transcription of the *lacZ*, *lacY*, and *lacA* genes.

